Task 1 - Discussion: Vandermonde Interpolation

1. Interpolation Polynomials for N = 3

Using the Vandermonde interpolation with $N = 3$ (i.e., 4 equidistant data points), the resulting polynomials are:

For $f_1(x) = \sin(2\pi x)$ on $[0, 1]$:

$$p_1(x) = 23.383\,x^3 - 35.074\,x^2 + 11.691\,x$$

For $f_2(x) = \frac{1}{1+x^2}$ on $[-3, 3]$:

$$p_2(x) = -0.05\,x^2 + 0.55$$

It is worth noting that for f_2, the odd-power coefficients vanish. This is expected since f_2 is an even function, and the equidistant nodes on $[-3, 3]$ are also symmetric, so the interpolation naturally produces an even polynomial.

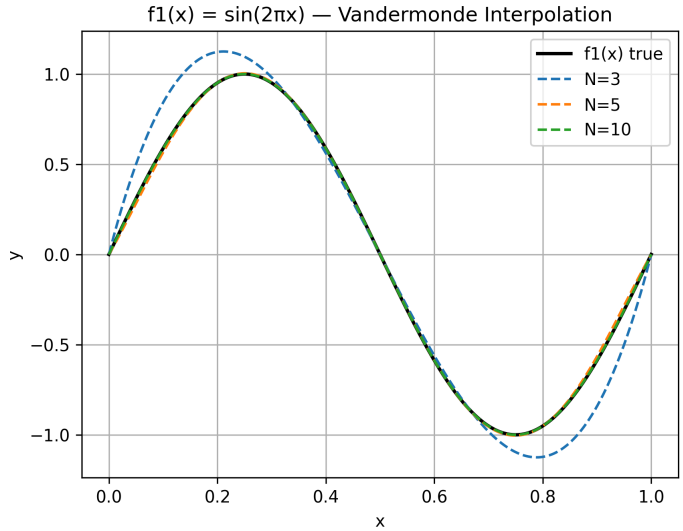
2. Approximation Quality for Increasing N

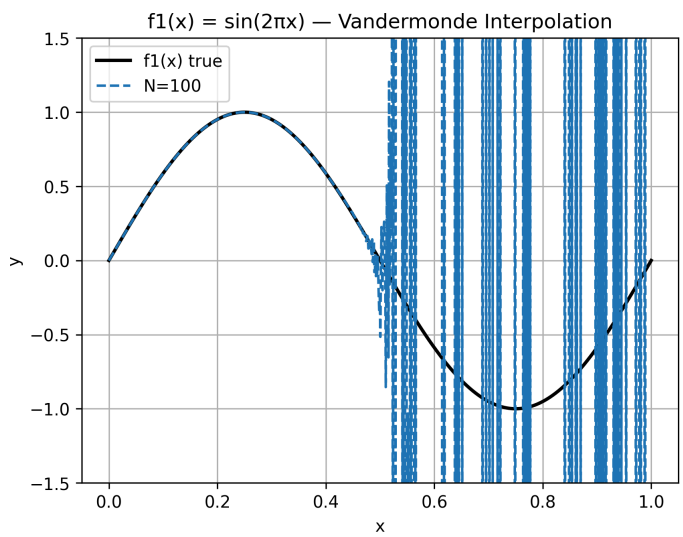
The code was run for $N = 3, 5, 10, 100$ and the interpolation polynomials were overlaid on the true functions.

$f_1(x)$: $\sin(2\pi x)$ on $[0, 1]$

- **N = 3:** The polynomial uses only 4 data points and cannot capture the peaks and troughs of the sine curve. Significant overshoot and undershoot are visible.
- **N = 5:** The fit improves substantially and the polynomial already follows the sine curve closely.
- **N = 10:** The interpolation is nearly indistinguishable from the true function.
- **N = 100:** The interpolation completely breaks down. While the polynomial roughly follows the sine curve for $x \in [0, 0.5]$, it exhibits **extreme high-frequency oscillations** for $x \in [0.5, 1.0]$, with amplitudes reaching the y-axis limits. The coefficients are on the order of 10^{36}, a direct

consequence of the catastrophically ill-conditioned Vandermonde matrix (Condition number $\kappa \approx 10^{20}$). The enormous coefficients cause severe **catastrophic cancellation** *(large terms that should nearly cancel each other fail to do so due to floating-point rounding)*, producing visible numerical noise in the evaluated polynomial.



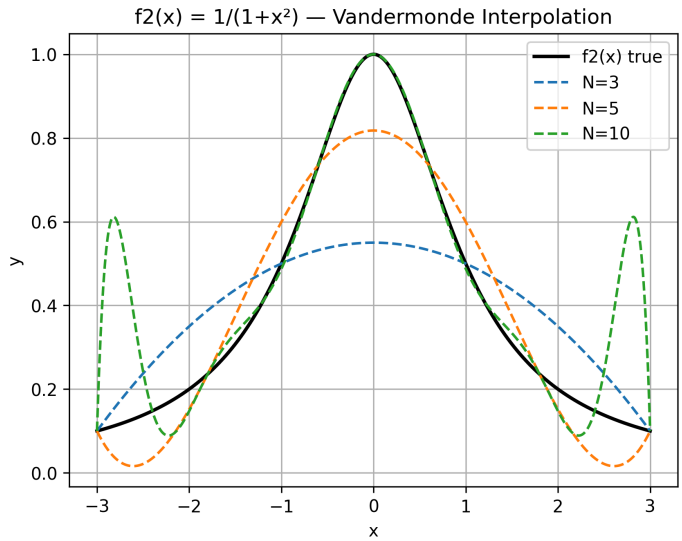


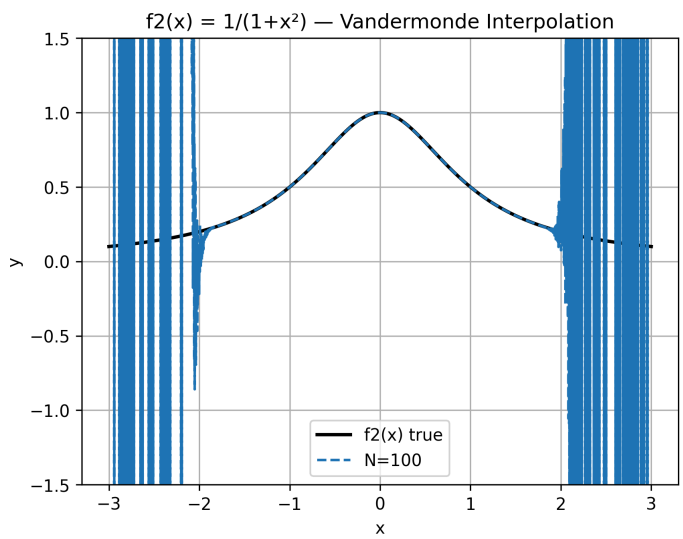
Conclusion for $f_1(x)$: Increasing N improves the approximation quality up to a point. For moderate N (5–10), the smooth nature of $\sin(2\pi x)$ on the short interval $[0, 1]$ allows polynomial interpolation to work effectively. However, at $N = 100$, the numerical instability of the Vandermonde matrix dominates, and the interpolation degrades drastically despite the function being smooth. This shows that even for well-behaved functions, the Vandermonde approach has a practical upper limit on N.

$f_2(x)$: $\frac{1}{1+x^2}$ on $[-3, 3]$

- **N = 3:** The polynomial completely fails to capture the peak at $x = 0$.
- **N = 5:** The general bell shape is captured, but the polynomial overshoots significantly, especially towards the domain boundaries.

- **N = 10:** The fit near the center improves, but large oscillations appear near the domain boundaries.
- **N = 100:** The oscillations dominate much of the domain and the interpolation is nearly useless.





Conclusion for $f_2(x)$: Increasing N does **not** improve the approximation. Instead, the oscillations near the boundaries grow worse. This is the **Runge phenomenon**, discussed in Section 3.

3. The Runge Phenomenon

The function $f_2(x) = \frac{1}{1+x^2}$ is the classic example introduced by Carl Runge in 1901 to demonstrate the failure of high-degree polynomial interpolation with equidistant nodes.

What happens: As N increases, the interpolation polynomial passes exactly through all $N+1$ data points. However, *between* the data points, especially near the boundaries of the domain, the

polynomial oscillates wildly. The interpolation error grows unboundedly near the edges even as it decreases near the center.

Why does this happen? The root cause lies in the distribution of the interpolation nodes. For equidistant nodes, the the interpolation error grows exponentially with N. This effect is particularly noticeable near the domain boundaries where equidistant nodes are sparsely distributed.

Why is $f_1(x)$ not affected?

1. The interval $[0, 1]$ is short, limiting the growth of high-degree terms.
2. $\sin(2\pi x)$ is an analytic, smooth function that is well-suited for polynomial approximation.

In contrast, f_2 is interpolated on the wider interval $[-3, 3]$, and its sharp central peak makes it inherently difficult for a single polynomial to approximate uniformly.

Possible remedy: Using **Chebyshev nodes** (which cluster near the boundaries) instead of equidistant nodes eliminates the Runge phenomenon by keeping the interpolation error growing only logarithmically with N. Piecewise interpolation techniques (Splines) or Lagrangian interpolation could also be suitable.

4. Simulation Time for Increasing N

The measured runtimes for constructing the Vandermonde matrix and solving the linear system are:

N	$f_1(x)$ time (s)	$f_2(x)$ time (s)
3	1.59e-04	1.79e-05
5	2.72e-05	1.24e-05
10	3.84e-05	1.41e-05
100	2.35e-02	2.69e-02

For small N (3, 5, 10), the runtimes are in the order of microseconds. However, the jump from $N = 10$ to $N = 100$ is significant: the runtime increases by approximately a factor of 1000.

This is consistent with the theoretical complexity:

- Constructing the Vandermonde matrix: $O(N^2)$
- Solving the linear system via `numpy.linalg.solve` : $O(N^3)$

The dominanting factor is the $O(N^3)$ complexity of the linalg solver. Since $\left(\frac{100}{10}\right)^3 = 1000$, the observed ~1000× increase aligns well with the expected cubic growth.

5. Condition Number of the Vandermonde Matrices

The condition numbers of the Vandermonde matrices were computed using `numpy.linalg.cond` :

N	κ for $f_1(x)$	κ for $f_2(x)$
3	9.89e+01	3.10e+01
5	4.92e+03	5.23e+02
10	1.16e+08	7.09e+05
100	**8.80e+20**	**2.14e+62**

The condition number κ measures how sensitive the solution of $V\mathbf{a} = \mathbf{y}$ is to perturbations in the input. A rule of thumb is that if $\kappa \approx 10^k$, roughly k floating-point digits are lost in the solution.

Observations:

- The condition number grows exponentially with N.
- At $N = 100$, $\kappa \approx 10^{62}$ for $f_2(x)$. Since `float64` provides only ~16 digits of precision, **all digits in the solution are corrupted**. The computed coefficients are numerically meaningless.
- Even for $f_1(x)$ at $N = 100$, $\kappa \approx 10^{20}$ means roughly 20 digits are lost; far exceeding the 16 available. The individual coefficients are comlplete numerical noise, and any reasonable-looking result from `np.polyval` is due to catastrophic cancellation (large terms nearly cancelling each other by chance).

Implication: The Vandermonde matrix approach is numerically unstable for large N. Even when the mathematical interpolation problem is well-posed, the numerical method fails because the matrix is too ill-conditioned for floating-point arithmetic to handle.